TOPIC 2 · PERSONALISATION, .RULES AND SKILLS FOR EXCEL
LAB 5 THEME · CUSTOM PARETO CHART SKILL

Lab 5 — Use the Built-in Chart Skill, then Create a Custom Pareto Skill

Maps to	LO2: distinguish account Personalisation, workbook .Rules and reusable Skills; configure, apply and verify each using water-service evidence.
Duration	30 minutes
Tools	Excel Copilot · Microsoft 365 Copilot web · OneDrive
Mock data	data/Northstar_Work_Orders.xlsx
Lab files	data/templates/water-service-pareto/SKILL.md (Reviewed Pareto Skill fallback)
Course	GenAI for Engineers · v2.3
Prompts	https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-5

Goal

Author water-service-pareto/SKILL.md with Copilot, install it in Excel and apply it to the same overdue work-order queue. Build a descending count chart plus cumulative-percentage line, then check a controlled rerun and a second grouping. Contribution analysis does not establish root cause or force an 80/20 split.

What You Will Do

Start with Excel's built-in @chart-design skill, then author water-service-pareto/SKILL.md with Copilot, install it in Excel and apply it to the same overdue work-order queue. Build a descending count chart plus cumulative-percentage line, then check a controlled rerun and a second grouping. Contribution analysis does not establish root cause or force an 80/20 split.

What You Will Build

water-service-pareto/SKILL.md, Northstar_Pareto.xlsx with Pareto_Job_Category and Pareto_Site, and a verified rerun record.

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.
- Lab 4 visible .Rules and clean source; assigned Excel Skill feature checked.

Data note

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Steps

Step 1. Open your rules-enabled workbook from Lab 4 or this lab's clean rules-enabled data/Northstar_Work_Orders.xlsx. Save as Northstar_Pareto.xlsx. Read the expected Pareto controls in Test It and the supplied Skill template before drafting your own file. Work only with synthetic data.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
For tblWorkOrders at Control!B2, All sites, count overdue backlog by Job_Category: Status
neither Closed nor Cancelled and Due_Date strictly before the reporting date. Show category
counts and total before charting. Preserve all records.
```

Step 2. Built-in skill first. In Excel Copilot open Settings (...) > Manage plugins & skills. Under Visualization turn on chart-design (it is off by default) and confirm it is enabled. Ask @chart-design for a quick chart of the same overdue counts. Note what the built-in skill gives you (chart design guidance) and what it does not know (your overdue population, tie order, cumulative line and named output) - that is why you build the custom water-service-pareto Skill next. Keep the demo sheet or undo it.

PROMPT / CHECK — copy and adapt

```
EXCEL - ALLOW EDITING:
@chart-design Chart the overdue backlog by Job_Category from tblWorkOrders at Control!B2 as a
bar chart, longest bar first, on a new sheet named ChartDesign_Demo. Use a finding-style
title, axis titles and data labels. Do not change source data.
```

Step 3. In Microsoft 365 Copilot on the web at https://copilot.cloud.microsoft, provide this self-contained specification. Copy the complete response into a plain-text editor and save UTF-8 SKILL.md inside a matching water-service-pareto folder. Compare it to data/templates/water-service-pareto/SKILL.md. If short of time, use the reviewed supplied file and record that choice.

PROMPT / CHECK — copy and adapt

```
MICROSOFT 365 COPILOT WEB:
Write a complete SKILL.md named water-service-pareto for a recurring water-pipe service
report. Require YAML name and description, actual source table, reporting date, scope, count
metric and grouping column. Source tblWorkOrders has Work_Order_ID, Job_Category, Site, Status
and Due_Date. Overdue backlog excludes Closed/Cancelled and has Due_Date before the supplied
date. Read visible workbook .Rules; preserve source IDs/statuses/owners and unrelated sheets.
Count by grouping, sort count descending then label alphabetically for ties, and create
formula-backed count, cumulative count, share and cumulative percentage. Build an editable
combo chart: count columns and cumulative line on a 0%-100% right axis. Output only
Pareto_<grouping column>; ask before replacing the matching output, rerun without duplicates,
and leave zero-total percentages N/A. Verify counts, stable ties and final 100%; do not infer
causes or force 80/20. Return the entire file.
```

Step 4. In Excel use Copilot Settings (...) > Manage plugins & skills > Custom skills > Create OneDrive folder if needed > Open skills folder. Copy water-service-pareto/SKILL.md into that app-created location, Refresh and enable it. Verify the name using Choose plugins & skills or @mention. If unavailable, paste the complete reviewed Skill before each request, remove @water-service-pareto and record Manual practice - native Skill unavailable. Log web authoring manually.

Step 5. Request a plan before building. Confirm the eligible total 50 and the five ordered category counts. After review, apply the Skill. Inspect Pareto_Job_Category, formulas, axes and source preservation; copying a static image does not meet the editable-chart goal.

PROMPT / CHECK — copy and adapt

```
EXCEL - PLAN:
@water-service-pareto Plan an overdue-backlog Pareto Chart by Job_Category from tblWorkOrders
at 2026-09-30, All sites. Follow visible .Rules. Explain summary formulas, descending counts,
alphabetical ties, cumulative percentages and the named output to create. Do not edit yet.

EXCEL - ALLOW EDITING:
@water-service-pareto Apply the reviewed Job_Category Pareto plan. Create Pareto_Job_Category
with a formula-backed summary and editable count-column/cumulative-percent-line combo chart.
Keep source data and unrelated sheets unchanged.
```

Step 6. Check counts against the supplied controls. Verify cumulative percentages 36%,60%,76%,92%,100% and two equal8-count categories in alphabetical order. The first four categories reach at least80%; do not call this proof that20% of categories cause80% of failures. If combo chart UI is absent, select the summary and use Insert > Combo > Custom Combination: Count clustered column, Cumulative_Percent line on secondary axis, right bounds 0 and 1; label both axes. Record a missing feature honestly.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
Verify Pareto_Job_Category against eligible source rows. Check total 50, descending counts,
alphabetical tie order, cumulative percentages 36/60/76/92/100%, final 100% and labelled dual
axes. State which category set first reaches 80%, and why this is contribution rather than a
causal diagnosis.
```

Step 7. Save a sandbox copy, rerun the same grouping after approving replacement of only Pareto_Job_Category, and compare source IDs/statuses/owners and sheet/chart counts. Then apply the Skill to Site to produce a distinct Pareto_Site output. Keep both reviewed summaries for later decision evidence.

PROMPT / CHECK — copy and adapt

```
EXCEL - PLAN:
@water-service-pareto Refresh the existing Pareto_Job_Category for the same source, date and
All scope. Identify exactly what will be replaced and ask before replacement. Preserve every
unrelated sheet and do not duplicate charts.

EXCEL - ALLOW EDITING:
@water-service-pareto After the approved category rerun, create a separate overdue-backlog
Pareto_Site for the same source/date/scope. Preserve Pareto_Job_Category and reconcile the new
summary to 50 overdue orders.
```

Sample Prompts

Use the app labelled for each prompt (or adapt it). Name the table and columns, state the output, and always verify the result. Use the linked prompts web page for one-click Copy buttons.

Copy-ready prompts: https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-5

#	Purpose	Prompt
1	Scope	State actual source, report date and selected scope before acting.
2	Verify	Explain the source/formula checks and preserve unknowns and original records.

Test It

Check	Expected result
Built-in skill	chart-design enabled under Visualization; @chart-design demo chart created without changing the source.
Eligible population	Overdue backlog50; no Closed or Cancelled records included.
Category counts	Pump Service=18; Meter Service=12; Leak Repair=8; Valve Repair=8; Pipe Replacement=4
Cumulative percentages	36%,60%,76%,92%,100%; Leak Repair precedes Valve Repair in the8-count tie.
Chart and rerun	Editable columns plus cumulative line; right axis0%-100%; matching output replaced only after review, without duplicates.
Source and interpretation	120 original records preserved; Site grouping also totals50; no invented root cause.

Checkpoint

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?